Peak Power: parameter audit of the investment model

Every parameter behind the "optimum investment approach", where it comes from, how it connects to the dataset, and how much the answer depends on it

Basis: symposium packet v6.0.1 ("V2"; supplied folder named packet_v6.6.1) in `data/Symposium_Packet` · **Models audited:** the browser engine (`peak-power/js/engine.js`, 384-plan library, profit layer) and the Python backend (`data/Symposium_Packet/team_app/engine.py`) · **Date:** 3 November 2026
Method: review of the packet files, both engines and the interface code; replay of all 384 library plans through the packet's official checker; one-at-a-time sensitivity runs with the app's own engine; unit and browser tests. **Prepared by** an AI assistant (Claude) at the team's request. This is a technical review, not a certification, and it does not assess whether the fictional benchmark resembles any real project.

1. Summary

The model chooses a power portfolio and an opening month for a fictional 75 MW Texas data-center campus by minimizing ten-year **decision cost** (capacity charges, contract payments, fuel, merchant power and the cost of moving or pausing work), optionally subject to a carbon limit. The calculation itself is sound and reproducible. The word "optimum" is the weak point: it means **the cheapest of a limited list of candidates under rule-based dispatch**, not a proven global optimum.

What holds up. All 384 browser plans pass the packet's official checker, and cost, CO_2 and clean share agree with it to within 0.05%. Packet data is hash-locked by the checker. The scripts that build the data are deterministic. 18 JavaScript and 6 Python unit tests and the browser tests pass.

What a reader should know before relying on a number.

- **The preferred mix is fragile around one price.** The recommended plan (150 MW wind + 50 MW solar + 25 MW gas turbine) loses to a 100 MW wind + 50 MW turbine plan if the wind contract price rises above about **$51.6/MWh**, only 15% above the packet's $45.
- **Wind is bought on a pay-for-available basis.** About 20% of contracted wind energy is paid for but not used. Extra wind output therefore *raises* cost (+15% output gives +4.2%).
- **Profit figures come from tenant prices the team invented**, not from the dataset. They are undiscounted and are not an NPV.
- **Battery rules use fixed dollar thresholds** while the price path escalates 5% a year, so the battery's charging window shrinks from 61% of hours in year 1 to 22% in year 10.
- **Several constants are copied into the code** instead of read from the data files (details in section 8), so a changed packet could be silently mis-modelled.

2. What the model decides

1 · Dataset	2 · Inputs	3 · Dispatch	4 · Accounting	5 · Selection
Packet files: hourly reference year, tiers, stress events, resource catalogue, grid schedule, price path.	87,600-hour timeline: the 2023 year repeated 10×, demand ramp, escalated prices, grid limit.	Hour by hour: contracts, grid, gas, battery, then relocate or pause work if allowed.	Capacity charge + contracts + fuel + merchant + response cost; CO_2; clean share.	Cheapest feasible candidate (or lowest-carbon). Then the profit layer ranks opening months.

Element	Definition in the model
Decision variables	Opening month (14–109; plans cover 14, 16, 18, 20, 22, 24, 27, 30); contract MW (wind ≤ 200, solar ≤ 150, nuclear ≤ 25); onsite MW (gas turbine in 25 MW blocks ≤ 100, gas engine ≤ 60, fuel cell ≤ 30, battery ≤ 53); flexibility policy (what work may move or pause during stress events).
Objective	Minimize **total decision cost** over 120 months (cost mode), or minimize combined CO_2 with cost as the tie-break (carbon mode). The packet states that preferences and weights are the team's to design.
Hard constraints	Hourly energy balance; resource availability dates; shared grid-import limit; capacity limits and block sizes; 8-acre land limit; battery power, energy, efficiency, cycle limit and zero start/end charge; Tier 0 always served locally; event eligibility and response limits; remote capacity 12 MW.
Search method	**Not an optimizer.** Browser: for each opening month × contract mix × battery (0 or 30 MW) × policy × objective, 14 candidate gas packages are simulated and the best feasible one is kept (384 plans). Python: 14 base portfolios × 4 battery sizes (0, 15, 30, 53 MW) = 56 candidates per policy and objective.
What "investment" adds	The prototype's profit layer multiplies served energy by assumed tenant rates and subtracts decision cost and unmet-demand penalties, to pick the opening month with the "highest 10-year profit". Those rates are **not in the packet**.

Cost is explicitly "a cumulative screening cost, not project NPV" (packet `case.json`): equipment is converted to annual charges at 8% real, there is no terminal value, no retirement and no second discounting.

3. The dataset and what each file feeds

Packet file	Content	Parameters it drives, with the dataset's own statistics
`reference_hourly.csv`	8,760 hours of 2023: wind and solar capacity factors, ERCOT LZ_SOUTH wholesale price, Henry Hub gas, load, Tier 0/1/2 demand, stress flags.	Wholesale price: mean $45.1, 95th percentile $87.4, min −$31.8, max $4,837/MWh; 185 negative-price hours, 169 above $200, 45 above $1,000. Wind capacity factor mean 0.326 (only 0.155 in stress hours); solar mean 0.212. Gas Henry Hub mean $2.54/MMBtu. Load mean 69.0 MW (min 63.0, max 75.0).
`timeline_default.csv`	The 87,600-hour timeline for opening month 20, already escalated.	Used by the official checker; other opening months are rebuilt with `prepare_inputs()` (Python) or `buildInputs()` (browser).
`resource_options.csv`	Seven offers: CAPEX, fixed O&M, life, energy price, heat rate, CO_2, availability month, block and maximum MW, land per MW.	Capacity limits, cost accounting and the land constraint (section 4D). The browser catalogue is regenerated from this file by `scripts/build_reference_data.py`.
`case.json`	Site, horizon, discount rate, grid schedule, storage rule, ramp, land.	Peak 75 MW, load factor 0.92, 120 months × 730 h, 8-acre limit, 8% real capital rate, grid limit 0 → 25 → 50 → 75 MW at months 0/18/24/30.
`response_parameters.json`	Demand-flexibility rules and prices.	Tier shares, 10 MW protected floor, 12 MW remote capacity, ×1.05 energy multiplier, $70 + $50 relocation, $1,000/MWh interruption, 60% stress import cap, 7-hour maximum event. In the data: **7 events, 32 stress hours**, mean wholesale price in those hours **$1,227/MWh**.
`stress_events.csv`	Seven events, 24 Aug – 7 Sep 2023.	Defines when relocation or pausing is allowed; the same events repeat in each of the ten years.
`price_trajectory.json`	Real annual escalation rates.	Wholesale +5%/yr, Henry Hub +3%/yr, grid CO_2 −3%/yr; year-10 multipliers 1.629, 1.344, 0.737.
`commissioning_ramp.csv`, `grid_connection_schedule.csv`	Demand ramp and import-limit steps.	One third of demand at opening, +1/15 per month, full at +10 months.
`assumptions_register.csv`, `source_register.csv`	Every assumption with its class (CASE assumption, CASE offer) and rationale.	All commercial offers and response parameters are labelled **CASE**: fictional benchmark assumptions, not quotes or observations.

> **Data provenance in one line.** Hourly shapes are historical proxies (2023 ERCOT) repeated ten times; prices, offers, response parameters and the site are benchmark assumptions. The packet itself says the result is a screening scenario, not a forecast.

4. Parameter register

Basis tags: Data value taken from a packet data file · Case packet "CASE" assumption or offer (fictional, documented) · Team team assumption, not in the packet · Code constant written into the code. "Where used" names the file or function. The last column links each parameter to the dataset and records audit notes.

A. Site, timeline and demand

Parameter	Value	Basis	Where used	Link to dataset / audit note
Mature demand	75 MW peak; load factor 0.92; 604,440 MWh/yr	Data	All dispatch	Matches `reference_hourly.csv` `load_mw`: mean 69.0 MW, max 75.0 MW.
Horizon and clock	120 months × 730 h = 87,600 h; reference year repeated 10×	Case	Timeline builders	No year-to-year variability, so the 32 stress hours and their prices recur identically each year.
Opening month	Default 20; allowed 14–109; campus ready month 14; full-load target month 30	Case	Decision variable	Browser library covers 14, 16, 18, 20, 22, 24, 27, 30. Opening month 20 is the packet default.
Demand ramp	1/3 of demand at opening, +1/15 per month, full at +10	Case Code	`commissioning_ramp.csv`; `engine.js` `1/3 + k/15`	The code's formula equals the 11 rows of the ramp file. It is a copy, not a read of the file.
Demand tiers	Tier 0 protected (mean 44.2 MW); Tier 1 relocatable (24% base, mean 16.5, max 20.7 MW); Tier 2 interruptible (12% base, mean 8.3, max 11.7 MW)	Case	Flexibility policies	Tier columns are in the hourly file. Tier shares are "proposed bounded benchmark", not measured practice.
Protected floor	10 MW	Case	Tier 0	Essential load always served locally.
Onsite land	8 acres; per MW: battery 0.15, turbine 0.10, engine 0.05, fuel cell 0.05	Case	Capacity limit	Footprints are in `resource_options.csv`. The browser repeats them in `js/app/core.js` (Code). At 0.15 acre/MW a 53 MW battery fills the site alone.

B. Grid access and stress events

Parameter	Value	Basis	Where used	Link to dataset / audit note
Import limit schedule	0 MW → 25 (m18) → 50 (m24) → 75 (m30); shared by merchant power and all offsite contracts	Data Code	`case.json`; `engine.js`	Python reads the file. The browser has the months and MW written into `buildInputs()` and `planning.js`.
Grid delay	Months, default 0	Case	Sensitivity	Shifts all positive milestones; other resource dates stay fixed. +6 months adds 2.1% to cost (section 7).

Parameter	Value	Basis	Where used	Link to dataset / audit note
Stress import cap	60% of the ordinary limit	Case	Stress hours	In the data: 32 hours in 7 events. Mean wholesale price in those hours is $1,227/MWh against $45.1 for the year.
Event rules	Maximum 7 h; 30 minutes' notice; relocation and pausing only inside events	Case	Eligibility masks	Browser relocates in every stress hour without a notice check; all dataset events carry 30 minutes, so results agree.

C. Prices, escalation and emissions factors

Parameter	Value	Basis	Where used	Link to dataset / audit note
Wholesale power price	LZ_SOUTH 2023 hourly + $20/MWh service adder	Data Case	Merchant cost	Mean $45.1, max $4,837. LZ_SOUTH is a regional proxy, not the campus's settlement point. Negative prices are credited.
Wholesale escalation	+5% a year real; year-10 multiplier 1.629	Case Code	`price_trajectory.json`; `engine.js` `Math.pow(1.05, …)`	Mean delivered price rises from about $66 to $92/MWh by year 10. Hard-coded in the browser.
Gas price	Henry Hub (mean $2.54) × (1.03)^years + $1/MMBtu adder	Data Case Code	Fuel cost	Escalation 3% a year; contract and equipment prices are not escalated.
Grid CO_2 factor	0.3329 t/MWh (733.9 lb/MWh eGRID), −3% a year	Data Case Code	Emissions	The remote receiving site uses the same factor. Annual-average screening factor, not marginal.
Gas CO_2 factors	Turbine 0.501, engine 0.449, fuel cell 0.345 t/MWh	Data	Emissions	Heat rate × 117 lb CO_2/MMBtu. Wind, solar and nuclear are zero operational CO_2 (not lifecycle).
Clean-energy threshold	100 kg CO_2/MWh (illustrative)	Case Code	`plant-review.js` `LIMDEF`	Of the 384 library plans, 288 are at or under 100 kg/MWh.

D. Resource catalogue (all seven offers, from `resource_options.csv`)

Resource	Kind	Block / max MW	Available (month)	CAPEX $/kW	Annual equip. cost $/kW-yr*	Energy price	Heat rate MMBtu/MWh	CO_2 t/MWh	Var. cost $/MWh**	Land ac/MW
utility_solar	variable	5 / 150	24	—	—	$40/MWh pay-for-available	—	0	—	0
wind	variable	5 / 200	12	—	—	$45/MWh pay-for-available	—	0	—	0

Resource	Kind	Block / max MW	Available (month)	CAPEX $/kW	Annual equip. cost $/kW-yr*	Energy price	Heat rate MMBtu/MWh	CO_2 t/MWh	Var. cost $/MWh**	Land ac/MW
battery_4h	storage	1 / 53	14	1,220	183	$0	—	0	—	0.15
gas_turbine	dispatchable	25 / 100	18	1,606	160	$5.7/MWh var. O&M	9.447	0.501	$39.1	0.1
nuclear	firm	5 / 25	60	—	—	$95/MWh pay-for-available	—	0	—	0
gas_engine	dispatchable	5 / 60	16	2,400	269	$12/MWh var. O&M	8.454	0.449	$41.9	0.05
fuel_cell	dispatchable	5 / 30	14	4,000	617	$5/MWh var. O&M	6.5	0.345	$28.0	0.05

* Annual equipment cost = CAPEX × CRF + fixed O&M, with CRF = r / (1 − (1 + r)$^{-life}$) at r = 8% (CRF 0.0937 for 25 years, 0.1019 for 20, 0.1168 for 15). ** Variable cost at the dataset's mean gas price ($2.54 + $1 adder, year-0 prices) = variable O&M + heat rate × delivered gas. For comparison the mean merchant price in year 1 is about $66/MWh, so onsite gas is cheaper than the grid on average and the grid wins mainly in cheap hours. All rows are Case : fictional offers. Nuclear is available only from month 60.

E. Battery (V2 rules)

Parameter	Value	Basis	Where used	Link to dataset / audit note
Size and cost	1–53 MW; 4 h; CAPEX $1,220/kW (was $1,744); fixed O&M $40/kW-yr; life 15 y	Case	Catalogue	CAPEX reflects a 30% cut from the previous packet; 53 MW is the land limit. Annual equipment cost ≈ $183/kW-yr.
Efficiency and cycles	Round trip 0.85 (one way √0.85); at most 365 equivalent cycles a year; SOC starts and ends at 0	Case	Checker, both engines	Both engines keep at most one full cycle a day.
Charging rule	From any source (grid, offsite contracts, onsite gas) inside the grid cap; behind the meter; no export, resale or ancillary revenue	Case	`case.json` storage rule	V2 change. The only value the battery can earn is avoided purchases.
Clean credit	Discharge earns clean credit in proportion to the year's clean share of charging	Case	Checker; `O.credit` in the browser	Grid-charged energy earns no clean credit.
Discharge / charge thresholds (browser)	Discharge above $100/MWh; charge from the grid below $45/MWh; hold window 72 h; end-of-horizon flush 96 h	Team Code	`engine.js` `ARB, CHARGE_PX, HOLD, FLUSH`	Fixed nominal dollars against a price path that escalates 5% a year: hours below $45 fall from 61% (year 1) to 22% (year 10), hours above $100 rise from 5.7% to 10.8% (figure 1).
Dispatch rule (Python)	Greedy daily arbitrage: pair the cheapest charging hour with a later hour whose avoided cost exceeds the round-trip loss	Team	`engine.py` `_battery_shift`	One cycle a day at most; does not help avoid unserved demand. A heuristic, not an optimizer.

F. Demand flexibility

Parameter	Value	Basis	Where used	Link to dataset / audit note
Remote capacity	12 MW electricity (11.43 MW local equivalent), × 1.05 energy	Case Code	`response_parameters.json`; `REMOTE_LOCAL`	Tier 1 relocation is limited to 11.43 MW in any hour (data: Tier 1 max 20.7 MW).
Relocation cost	1.05 × $70 + $50 = $123.50 per local MWh	Case Code	`REL_PRICE`	Written as a formula in the browser; the Python side reads the JSON.
Interruption consequence	$1,000/MWh for paused Tier 2 work	Case Code	`INT_PRICE`, `UNMET`	Fictional "1 USD per kWh", not observed revenue loss.
Policies	Browser: never / last resort / when cheaper. Python: strict / relocate / flexible	Team	Dispatch	Names differ and so does meaning: the browser's "never" still relocates Tier 1 work and only forbids pausing; Python's "strict" forbids both.
"When cheaper" trigger	Pause Tier 2 work when the price exceeds $1,000/MWh net of gas cost	Team Code	`ival` default 1000	Ties pausing to the penalty value.

G. Cost accounting and search settings

Parameter	Value	Basis	Where used	Link to dataset / audit note
Capital-charge rate	8% real	Data Code	`case.json`; `crf(0.08, …)`	Converts CAPEX to a yearly charge. Hard-coded in the browser and in `planning.js`. Moving 6% to 10% changes cost by ±1.9%.
Start and commitment	Charges begin at max(opening, availability); capacity stays committed through month 120; no resale, retirement or terminal value	Case	Accounting	An early-available resource is paid for even if later supply displaces it.
Contracts	Price × MW × capacity factor on all available hours, used or curtailed	Case	Accounting	This makes unused wind a pure cost; see section 7.
Candidate lists	Browser: gas packages (turbine, engine, fuel cell MW) [0,5,0] [0,10,0] [0,15,0] [0,25,0] [0,35,0] [0,0,30] [0,15,30] [25,0,0] [25,5,0] [25,10,0] [25,30,0] [25,10,30] [50,0,0] [50,10,0]; contract mixes W100, W150S50, W200S100, W200S100N25. Python: 14 base portfolios × battery 0/15/30/53	Team	`build_plan_library.mjs`; `engine.py`	Anything outside these lists is invisible to the "optimum". Mixes with no solar below 150 MW, or with larger gas, are not tried.

Parameter	Value	Basis	Where used	Link to dataset / audit note
Selection rule	Cost mode: lowest total cost, then lowest CO_2. Carbon mode: lowest CO_2, then lowest cost. Feasible = no unserved hour (and no pausing under "never")	Team	Both engines	No weighting between cost and carbon. The user's carbon cap is applied afterwards in the interface.

H. The investment (profit) layer in the prototype

Parameter	Value	Basis	Where used	Link to dataset / audit note
Revenue model	Default "lease": $/kW-month by tier. Alternatives: lease + power pass-through, all-in $/MWh, combined	Team	`settings.js` `econ()`	Not in the packet. Served MWh by tier come from the dataset's Tier 0/1/2 columns; the prices are invented.
Lease rate cards	Low $140 / $110 / $60; mid $180 / $150 / $100; high $230 / $190 / $120 per kW-month (Tier 0 / 1 / 2). Default mid	Team	`RPRE`	Converted to $/MWh by × 1000 / 730.
Other tenant prices	Pass-through markup 0 / 5 / 10 / 20% (default 10%); all-in price $225–350/MWh (default $275)	Team	`RM`	The "best opening month" is the one maximizing revenue minus decision cost minus the $1,000/MWh unmet penalty.
Extensions	Demand growth, weather scaling (wind 0.8–1.15, solar 0.85–1.1), grid scenarios, rented generator at $350/MWh and 0.74 t CO_2/MWh	Team	`settings.js`, `engine.js`	Labelled as extensions in the interface.

5. How the parameters bite

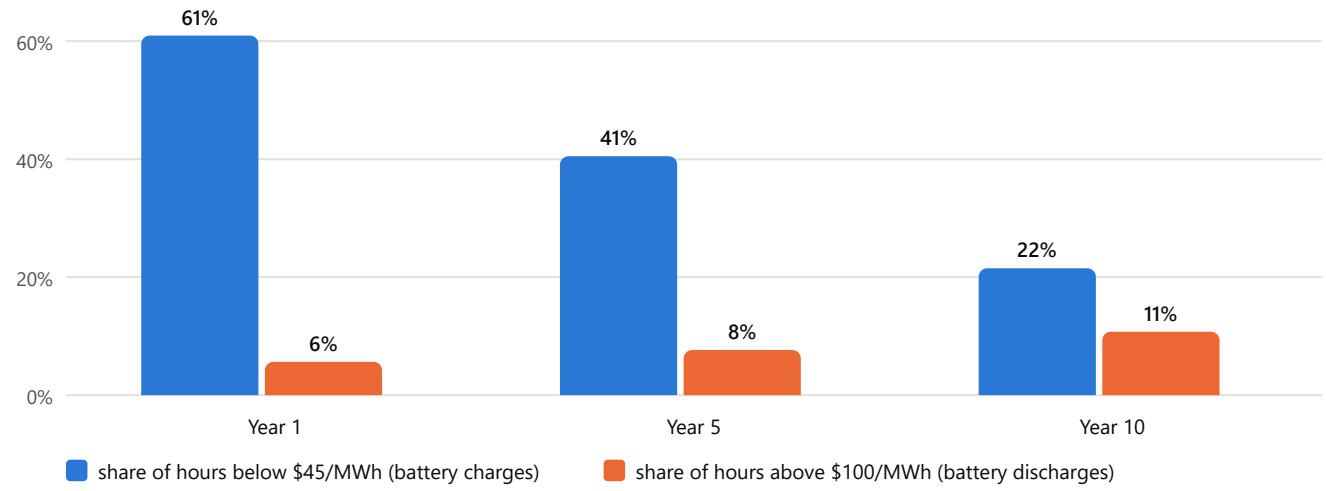


Figure 1. Battery price zones against the dataset's price path (browser engine). The thresholds ($45 charge, $100 discharge) are fixed, but the dataset's prices are escalated 5% a year, so the charging zone shrinks and the discharge zone grows. Source: `reference_hourly.csv` prices × the escalation in `price_trajectory.json`.

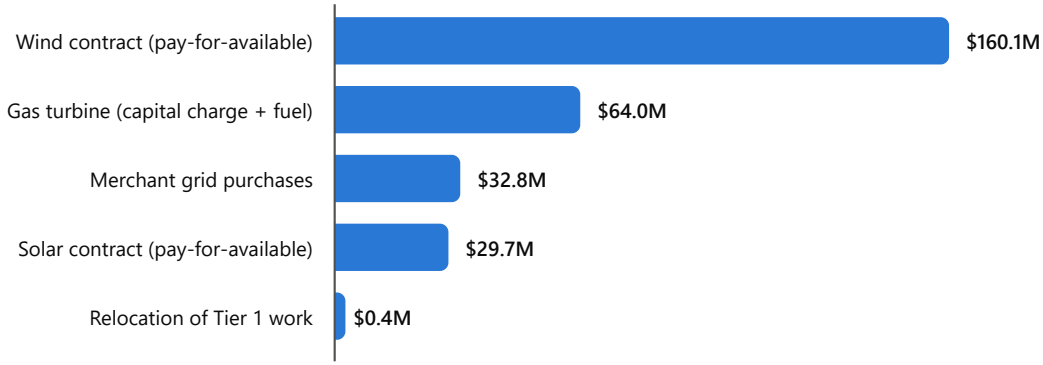


Figure 2. Ten-year cost structure of the browser's recommended plan for opening month 20 (150 MW wind, 50 MW solar, 25 MW gas turbine; total $287.0M). Wind contract payments are 56% of cost. Of 3.86 TWh of wind energy made available, 3.09 TWh (80%) is used; the other 20% is paid for and curtailed. Solar use is 93%.

6. What the model currently recommends

Engine and setting	Portfolio (MW)	10-year cost	Carbon	Notes
Python, cost, any policy, opening month 20	Turbine 25, engine 5, wind 100, solar 50, no battery	$283.1M	754 kt; 155.4 kg/MWh; clean 62.6%	Same portfolio under all three flexibility policies: it is feasible without relocating or pausing.
Python, carbon objective	Turbine 25, fuel cell 5, wind 150, solar 50, **battery 30**	$399.5M	294 kt; 60.6 kg/MWh	Battery charged from surplus clean energy; under the 100 kg threshold.

Engine and setting	Portfolio (MW)	10-year cost	Carbon	Notes
Browser, cost, "when cheaper", month 20: W100 + GT50	Wind 100, turbine 50	$294.8M	231 kg/MWh	Candidate mixes, each with its cheapest feasible gas package.
… W150S50 + GT25 (**cheapest**)	Wind 150, solar 50, turbine 25	**$287.0M**	94 kg/MWh	
… W200S100 + GT25	Wind 200, solar 100, turbine 25	$337.4M	48 kg/MWh	
… W200S100N25 + GT25	Adds 25 MW nuclear	$426.1M	29 kg/MWh	

The two engines disagree on the cheapest portfolio ($283.1M vs $287.0M, different mixes) because their candidate lists, dispatch rules and policies differ. That is expected, but it means the app should not present either as *the* optimum. The 384-plan browser library ranges from $258.2M to $665.2M. The previous packet's best portfolio cost $259.4M; most of the V2 increase comes from the wind price rising from $35 to $45/MWh.

7. Sensitivity of the recommended plan

Each parameter was changed one at a time with the browser's own engine, holding the recommended plan fixed (opening month 20, "when cheaper", cost objective). The plan stayed feasible in every case.

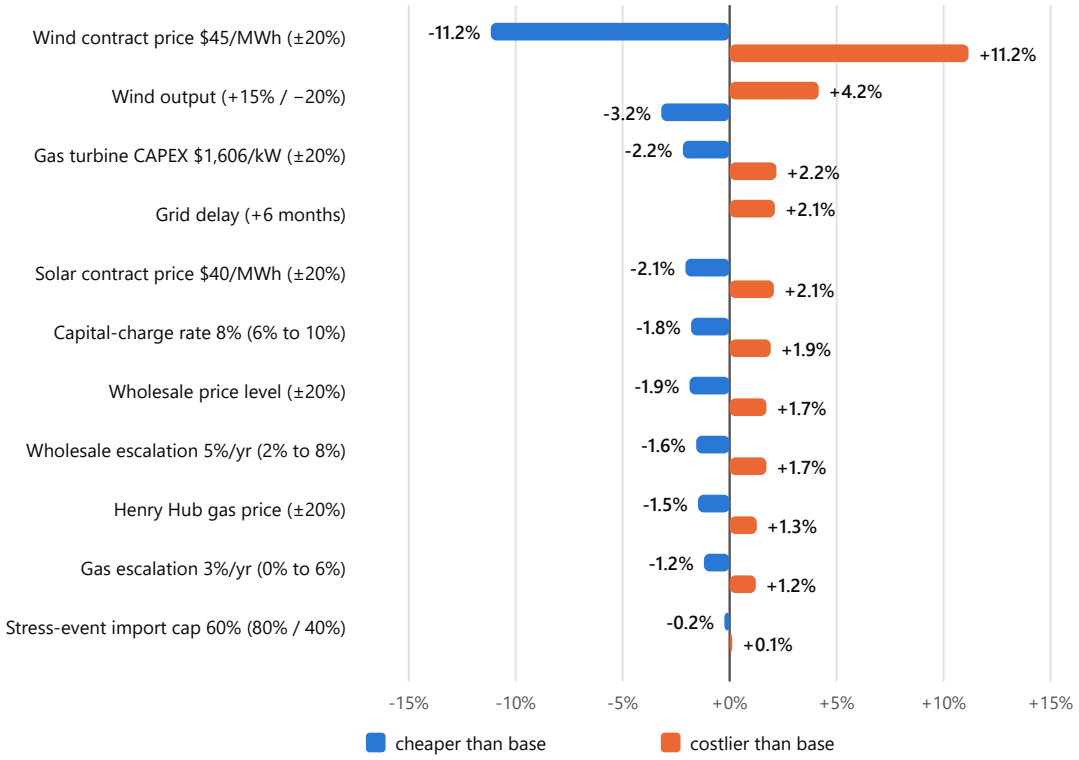


Figure 3. Change in ten-year cost of the recommended plan ($287.0M) when one parameter moves. Each row shows the effect of each tested value.

- **Wind price dominates:** ±20% moves cost by ±11.2%, about five times the effect of any other price. At $51.6/MWh the cheapest mix switches from W150S50 + GT25 to W100 + GT50 (cost crossover found by interpolating runs at $45–54).
- **Wind output cuts the other way:** 15% more wind output costs 4.2% more, because the extra energy is paid for and mostly curtailed.
- **Escalation, gas price, discount rate, equipment cost and grid delay** each move cost by 2.2% or less. They are worth stating but do not change the ranking.

- Only the wind price (and wind output) changed which mix is cheapest; the stress import cap has almost no effect on cost (−0.2% / +0.1%), because the recommended plan has enough local gas. Demand-tier shares were not varied.

8. Audit findings and recommendations

#	Rating	Finding	Evidence	Recommendation
1	**High**	"Optimum" is overstated. It is the cheapest of a fixed candidate list under heuristic dispatch.	Candidate lists in section 4G; no solver; the packet says "no optimal portfolio results". Interface text says the highest-profit month "wins".	Say "lowest-cost among N tested plans". If time allows, add a continuous search around the winner (contract MW in 5 MW steps) and report the gap.
2	**High**	The profit ranking rests on tenant prices the team chose, not on the dataset.	Lease rates, markups and all-in prices are in `settings.js`; none appear in the packet. Profit is undiscounted and unlevered.	Label profit as illustrative; show the opening-month ranking under the low, mid and high rate cards side by side; keep decision cost as the headline.
3	**High**	The mix decision flips at a small price change.	Wind $51.6/MWh crossover (+15%); wind price moves cost ±11.2% (figure 3).	Present the wind price as the key assumption, with the break-even price next to the recommendation.
4	**Medium**	Constants are copied into the code instead of read from the data files.	Escalation 1.05 / 1.03 / 0.97, ramp 1/3 + k/15, grid steps 18/24/30 and 25/50/75 MW, relocation $123.50, interruption $1,000, 12 MW remote, 8% rate, land per MW (`engine.js`, `core.js`, `planning.js`). All equal the current packet.	Generate these from the packet in `build_reference_data.py` (the catalogue and hourly series already are), or add a test that fails when the code and files differ.
5	**Medium**	Battery thresholds are fixed dollars against an escalating price path.	Hours below $45 fall from 61% to 22%; above $100 rise from 5.7% to 10.8% (figure 1).	Use thresholds relative to each year's price distribution, or an optimizer. Until then, treat the browser's battery results as conservative.
6	**Medium**	The two engines use different policy meanings and battery logic.	Browser "never" relocates; Python "strict" does not. Different candidate sets and a different cheapest portfolio (section 6).	Rename the policies to match, and state on the page which engine produced each figure.
7	**Medium**	Pay-for-available wind and solar can make extra capacity pure cost.	20% of wind energy is paid for and unused; +15% wind output raises cost 4.2%.	Show curtailed contract energy in the plant review so users see it; consider a candidate with 100–125 MW wind and more solar.
8	**Medium**	One repeated year carries ten years of stress.	32 stress hours at a mean $1,227/MWh recur every year; 45 hours above $1,000.	State that flexibility and battery values depend on these 32 hours; test a calmer year if one becomes available.
9	**Low**	The cost metric is a cumulative screening cost, not NPV or return.	`case.json` "cost_comparison"; no terminal value or retirement.	Keep the existing label; do not add ROI or payback.
10	**Low**	Battery library covers one size (30 MW); Python tries 4 sizes.	Section 4G.	Mention in the interface that the battery option is a 30 MW example.
11	**Low**	Real-world constraints are outside scope.	`case.json` optional list: unit commitment, startup, ramping, derating, degradation, outages.	Keep the "screening, not a guarantee" wording.

#	Rating	Finding	Evidence	Recommendation
12	**Low**	Embedded hourly data had been rounded too coarsely for the checker (found and fixed in this review).	Browser dispatch failed the checker's 1e-5 MW tolerance until the series were rebuilt at 9 decimals.	None: `build_reference_data.py` now regenerates the series from the packet.

9. Verification performed and how to reproduce

Check	Result
All 384 browser plans replayed through the packet's official checker (physical rules and accounting)	384 of 384 pass; cost, CO_2 and clean share within 0.05% of the library values.
Python backend dispatch against the official checker	Passes for the default and carbon portfolios; the Python unit tests verify the V2 battery rules.
Browser engine unit tests	Balance, shared import cap, state of charge, power and cycle limits, grid charging, clean-credit identity, no change without a battery.
Browser end-to-end tests	Prototype runs steps 1–3 with the V2 data at 1400 and 390 px with no errors; the explorer and navigation tests pass.
Not tested	Optimality of any result; accuracy of the benchmark against any real site; real Open-Meteo calls (mocked in tests); a physical phone.

To reproduce: `python scripts/build_reference_data.py` (catalogue and hourly data) · `node scripts/build_plan_library.mjs` (384 plans, about 11 minutes) · `python scripts/verify_plan_library.py --all` (official checker, about 25 minutes) · `python data/Symposium_Packet/team_app/run.py --compare-policies` (Python outputs) · `node --test tests/unit/*.test.js` and `python -m unittest tests/unit/test_python_engine.py`.

Sensitivity method. The sensitivity runs use a copy of `engine.js` with the three escalation rates exposed; the file itself was not changed. The discount rate was varied by rescaling CAPEX by CRF(r) / CRF(8%), which leaves the charge formula exact. Variations are one at a time, so interactions are not captured.